

CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the textbook exercises: 20 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 7.1

Q1 Find the distance between each pair of points: (i) (2, 3), (4, 1) (ii) (-5, 7), (-1, 3) (iii) (a, b), (-a, -b).

SOLUTION

Use the distance formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

$$(i) \sqrt{(4 - 2)^2 + (1 - 3)^2} = \sqrt{4 + 4} = 2\sqrt{2}$$

$$(ii) \sqrt{(-1 + 5)^2 + (3 - 7)^2} = \sqrt{16 + 16} = 4\sqrt{2}$$

$$(iii) \sqrt{(-2a)^2 + (-2b)^2} = \sqrt{4a^2 + 4b^2} = 2\sqrt{a^2 + b^2}$$

Answer: (i) $2\sqrt{2}$ (ii) $4\sqrt{2}$ (iii) $2\sqrt{a^2 + b^2}$ units

Q2 Find the distance between (0, 0) and (36, 15). Hence find the distance between the two towns of Section 7.2 of the textbook, where town B is 36 km east and 15 km north of town A.

SOLUTION

$$\text{Distance} = \sqrt{36^2 + 15^2} = \sqrt{1296 + 225} = \sqrt{1521} = 39.$$

Taking town A as the origin, town B is the point (36, 15), so the towns are 39 km apart.

Answer: 39 units, so the towns are 39 km apart

Q3 Are the points (1, 5), (2, 3) and (-2, -11) collinear?

SOLUTION

$$\text{Call them } A, B \text{ and } C. AB = \sqrt{1 + 4} = \sqrt{5} \approx 2.24, BC = \sqrt{16 + 196} = \sqrt{212} \approx 14.56 \text{ and } AC = \sqrt{9 + 256} = \sqrt{265} \approx 16.28.$$

The two shorter distances add up to about 16.80, which is not equal to the longest distance 16.28.

Answer: no, the points are not collinear

Q4 Are (5, -2), (6, 4) and (7, -2) the vertices of an isosceles triangle?

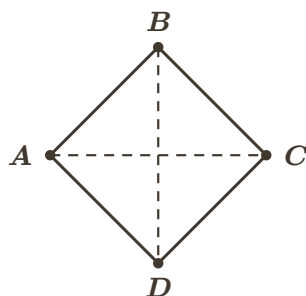
SOLUTION

$$\text{Call them } A, B \text{ and } C. AB = \sqrt{1 + 36} = \sqrt{37}, BC = \sqrt{1 + 36} = \sqrt{37} \text{ and } AC = \sqrt{4 + 0} = 2.$$

$AB = BC$, and the points are not collinear since $AB + AC \neq BC$.

Answer: yes, the triangle is isosceles with $AB = BC$

- Q5** Four friends sit at the points $A(3, 4)$, $B(6, 7)$, $C(9, 4)$ and $D(6, 1)$ of a classroom grid (read from the textbook figure). One student says $ABCD$ is a square and another disagrees. Who is correct?



$A(3, 4)$, $B(6, 7)$, $C(9, 4)$, $D(6, 1)$

SOLUTION

Sides: $AB = \sqrt{3^2 + 3^2} = \sqrt{18}$, $BC = \sqrt{3^2 + 3^2} = \sqrt{18}$, $CD = \sqrt{18}$ and $DA = \sqrt{18}$. All four sides are equal.

Diagonals: $AC = \sqrt{6^2 + 0^2} = 6$ and $BD = \sqrt{0^2 + 6^2} = 6$. The diagonals are equal.

A quadrilateral with four equal sides and equal diagonals is a square.

Answer: $ABCD$ is a square, so the student who said so is correct

- Q6** Name the type of quadrilateral formed, if any, by these points, with reasons: (i) $(-1, -2)$, $(1, 0)$, $(-1, 2)$, $(-3, 0)$ (ii) $(-3, 5)$, $(3, 1)$, $(0, 3)$, $(-1, -4)$ (iii) $(4, 5)$, $(7, 6)$, $(4, 3)$, $(1, 2)$.

SOLUTION

Call the points A, B, C, D in order.

(i) $AB = BC = CD = DA = \sqrt{4 + 4} = \sqrt{8}$. Diagonals: $AC = 4$ and $BD = 4$. All sides are equal and the diagonals are equal, so it is a square.

(ii) $AB = \sqrt{36 + 16} = 2\sqrt{13}$, $BC = \sqrt{9 + 4} = \sqrt{13}$ and $AC = \sqrt{9 + 4} = \sqrt{13}$. Since $AC + CB = AB$, the points A, C, B lie on one line, so no quadrilateral is formed.

(iii) $AB = \sqrt{10}$, $BC = \sqrt{18}$, $CD = \sqrt{10}$, $DA = \sqrt{18}$. Diagonals: $AC = 2$ and $BD = \sqrt{36 + 16} = \sqrt{52}$. Opposite sides are equal but the diagonals are not, so it is a parallelogram.

Answer: (i) a square (ii) no quadrilateral (iii) a parallelogram

- Q7** Find the point on the x -axis that is equidistant from $(2, -5)$ and $(-2, 9)$.

SOLUTION

Let the point be $(x, 0)$. Then $(x - 2)^2 + 25 = (x + 2)^2 + 81$.

$x^2 - 4x + 29 = x^2 + 4x + 85$ gives $-8x = 56$, so $x = -7$.

Answer: $(-7, 0)$

Q8 Find the values of y for which the distance between $P(2, -3)$ and $Q(10, y)$ is 10 units.

SOLUTION

$$(10 - 2)^2 + (y + 3)^2 = 100 \text{ gives } (y + 3)^2 = 36, \text{ so } y + 3 = \pm 6.$$

Answer: $y = 3$ or $y = -9$

Q9 $Q(0, 1)$ is equidistant from $P(5, -3)$ and $R(x, 6)$. Find the values of x , and the distances QR and PR .

SOLUTION

$$QP^2 = 25 + 16 = 41 \text{ and } QR^2 = x^2 + 25. \text{ Setting them equal, } x^2 = 16, \text{ so } x = \pm 4.$$

In both cases $QR = \sqrt{41}$.

$$\text{If } R = (4, 6): PR = \sqrt{1 + 81} = \sqrt{82}. \text{ If } R = (-4, 6): PR = \sqrt{81 + 81} = 9\sqrt{2}.$$

Answer: $x = \pm 4$, $QR = \sqrt{41}$, and $PR = \sqrt{82}$ or $9\sqrt{2}$

Q10 Find a relation between x and y such that (x, y) is equidistant from $(3, 6)$ and $(-3, 4)$.

SOLUTION

$$(x - 3)^2 + (y - 6)^2 = (x + 3)^2 + (y - 4)^2.$$

$$-6x - 12y + 45 = 6x - 8y + 25 \text{ gives } 12x + 4y - 20 = 0.$$

Answer: $3x + y - 5 = 0$

2 Exercise 7.2

Q1 Find the point that divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$.

SOLUTION

$$\text{Section formula: } \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right) = \left(\frac{2 \times 4 + 3 \times (-1)}{5}, \frac{2 \times (-3) + 3 \times 7}{5} \right).$$

Answer: $(1, 3)$

Q2 Find the points of trisection of the segment joining $(4, -1)$ and $(-2, -3)$.

SOLUTION

Let $A = (4, -1)$ and $B = (-2, -3)$. The first point P divides AB in the ratio $1 : 2$: $P =$

$$\left(\frac{-2 + 8}{3}, \frac{-3 - 2}{3} \right) = \left(2, -\frac{5}{3} \right).$$

The second point Q divides AB in the ratio $2 : 1$: $Q = \left(\frac{-4 + 4}{3}, \frac{-6 - 1}{3} \right) = \left(0, -\frac{7}{3} \right).$

Answer: $\left(2, -\frac{5}{3} \right)$ and $\left(0, -\frac{7}{3} \right)$

Q3 On a rectangular ground $ABCD$, lines are drawn 1 m apart, and 100 flower pots stand 1 m apart along AD . One student runs $\frac{1}{4}$ of AD along the 2nd line and posts a green flag. Another runs $\frac{1}{5}$ of AD along the 8th line and posts a red flag. Find the distance between the flags. Where should a blue flag go so that it is exactly halfway between them?

SOLUTION

Take A as the origin, AB as the x -axis and AD as the y -axis, with $AD = 100$ m.

Green flag: on the 2nd line at $\frac{1}{4} \times 100 = 25$ m, that is $(2, 25)$. Red flag: on the 8th line at $\frac{1}{5} \times 100 = 20$ m, that is $(8, 20)$.

Distance = $\sqrt{(8-2)^2 + (20-25)^2} = \sqrt{61}$ m.

Mid-point = $\left(\frac{2+8}{2}, \frac{25+20}{2}\right) = (5, 22.5)$.

Answer: $\sqrt{61}$ m apart. The blue flag goes on the 5th line, 22.5 m from AB .

Q4 In what ratio does $(-1, 6)$ divide the segment joining $(-3, 10)$ and $(6, -8)$?

SOLUTION

Let the ratio be $k : 1$. Then $\frac{6k-3}{k+1} = -1$ gives $7k = 2$, so $k = \frac{2}{7}$.

Check with y : $\frac{-8 \times 2 + 10 \times 7}{9} = 6$.

Answer: $2 : 7$

Q5 In what ratio does the x -axis divide the segment joining $A(1, -5)$ and $B(-4, 5)$? Find the point of division.

SOLUTION

Let the ratio be $k : 1$. The y -coordinate of the point is 0: $\frac{5k-5}{k+1} = 0$, so $k = 1$.

The point is the mid-point: $\left(\frac{1-4}{2}, 0\right) = \left(-\frac{3}{2}, 0\right)$.

Answer: $1 : 1$, at $\left(-\frac{3}{2}, 0\right)$

Q6 The points $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$, in that order, are the corners of a parallelogram. Find x and y .

SOLUTION

The diagonals of a parallelogram bisect each other, so the two diagonals have the same mid-point.

$\frac{1+x}{2} = \frac{4+3}{2}$ gives $x = 6$, and $\frac{2+6}{2} = \frac{y+5}{2}$ gives $y = 3$.

Answer: $x = 6$, $y = 3$

Q7 AB is a diameter of a circle with centre $(2, -3)$, and B is $(1, 4)$. Find A .

SOLUTION

The centre is the mid-point of AB . Let $A = (x, y)$: $\frac{x+1}{2} = 2$ gives $x = 3$, and $\frac{y+4}{2} = -3$ gives $y = -10$.

Answer: $A = (3, -10)$

Q8 A and B are $(-2, -2)$ and $(2, -4)$. Find the point P on the segment AB such that $AP = \frac{3}{7}AB$.

SOLUTION

$AP = \frac{3}{7}AB$ means $PB = \frac{4}{7}AB$, so P divides AB in the ratio $3 : 4$.

$$P = \left(\frac{3 \times 2 + 4 \times (-2)}{7}, \frac{3 \times (-4) + 4 \times (-2)}{7} \right) = \left(-\frac{2}{7}, -\frac{20}{7} \right).$$

Answer: $P = \left(-\frac{2}{7}, -\frac{20}{7} \right)$

Q9 Find the points that divide the segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

SOLUTION

The middle point is the mid-point of AB : $(0, 5)$.

The other two are the mid-points of each half: $\left(\frac{-2+0}{2}, \frac{2+5}{2} \right) = \left(-1, \frac{7}{2} \right)$ and $\left(\frac{0+2}{2}, \frac{5+8}{2} \right) = \left(1, \frac{13}{2} \right)$.

Answer: $\left(-1, \frac{7}{2} \right)$, $(0, 5)$ and $\left(1, \frac{13}{2} \right)$

Q10 Find the area of the rhombus with vertices $(3, 0)$, $(4, 5)$, $(-1, 4)$ and $(-2, -1)$ taken in order.

SOLUTION

Call the vertices A, B, C, D . Diagonals: $AC = \sqrt{(-4)^2 + 4^2} = 4\sqrt{2}$ and $BD = \sqrt{(-6)^2 + (-6)^2} = 6\sqrt{2}$.

$$\text{Area} = \frac{1}{2} \times AC \times BD = \frac{1}{2} \times 4\sqrt{2} \times 6\sqrt{2} = 24.$$

Answer: 24 square units

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